

Notes: Baker, Gibbs, and Holmstrom (QJE, 1994)

The Internal Economics of the Firm: Evidence from Personnel Data

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1 Big picture

- **Question.** What does the internal labour market of a large firm actually look like when one observes job titles, hierarchy, promotions, exits, salaries, and performance over many years?
- **Data contribution.** The paper analyzes roughly twenty years of personnel data from one large U.S. firm, covering 1969–1988 and tens of thousands of management employee-year observations.
- **Main object.** The paper reconstructs the firm’s internal hierarchy from actual title-to-title transitions, rather than relying on an organization chart or survey responses.
- **Core finding 1.** The hierarchy is simple, stable, and strongly ordered. Job titles can be grouped into eight levels, and movements across titles mostly follow a clear promotion ladder.
- **Core finding 2.** Careers have strong internal structure. Employees move through levels and titles in ways consistent with an internal labour market, not with a sequence of external spot-market matches.
- **Core finding 3.** Promotions matter for pay, but job level alone does not determine compensation. There is substantial salary dispersion within levels and titles.
- **Core finding 4.** Exit rates vary surprisingly little with tenure, level, salary, and performance. This weakens simple interpretations based on ports of exit or strongly firm-specific human capital.
- **Core finding 5.** There are promotion “fast tracks.” Employees who move quickly early in their career continue to move faster and to reach higher levels.
- **Core finding 6.** Pay policies include strong administrative structure. Raises are tied to levels, performance, promotions, and position in the within-title salary distribution.
- **Main interpretation.** The firm has many features of an internal labour market described by Doeringer and Piore, but several theoretical preconditions traditionally associated with internal labour markets are weak or absent.
- **Takeaway.** The firm is not a black box. Its internal labour market has a coherent hierarchy, promotion dynamics, and wage rules that can be studied quantitatively and that discipline theories of organizations.

2 Incremental contribution of the paper

2.1 Contribution relative to internal labour market theory

- Before this paper, much of the internal labour market literature was based on institutional descriptions and stylized facts rather than detailed longitudinal personnel records.
- Doeringer and Piore emphasized features such as ports of entry, internal mobility, job ladders, administrative wage rules, and long-term attachment.
- Baker, Gibbs, and Holmstrom provide a quantitative, firm-level test of these ideas using actual personnel data.
- The paper shows that many descriptive claims about internal labour markets are right: levels matter, careers follow ladders, promotions are central, and pay is administratively structured.
- At the same time, the paper challenges parts of the theory: ports of entry and exit are not as clean as the classic account suggests, and exit rates are not strongly tied to tenure, level, or salary.

Interpretation: The incremental contribution is not merely documenting that this firm has a hierarchy. The paper shows that one can use personnel data to evaluate internal labour market theory directly and to separate descriptive regularities from theoretical explanations.

2.2 Contribution relative to labour economics and personnel economics

- The paper is an early example of what became modern personnel economics: using within-firm administrative data to study wages, promotions, careers, and incentives.
- Rather than inferring firm behaviour from labour market surveys, the paper observes the internal allocation and pay rules of a single organization.
- It shows that job titles and hierarchical levels are economically meaningful state variables.
- It also shows that individual wage variation within levels is large enough that models based only on job-level wage rates are incomplete.

Interpretation: The paper helps move labour economics from a market-level view of wages toward a firm-internal view of careers and organizational rules.

2.3 Contribution relative to theories of promotions and asymmetric information

- Models by Waldman and others emphasize that promotions reveal information about ability and can affect careers through signalling.
- The paper documents that promotions are persistent and path-dependent: early promotion speed predicts later career progress.
- It also shows that performance ratings and salary position matter for raises and promotions, but not in a simple one-dimensional way.
- The evidence supports theories in which firms learn about worker ability over time, but it also reveals institutional wage rules that pure signalling models do not capture.

Interpretation: The incremental contribution is empirical discipline: theories of careers must explain the combination of fast tracks, level structure, within-level wage dispersion, promotion premia, and weak exit patterns.

2.4 Contribution relative to the companion paper

- This paper focuses on the internal structure of jobs, movements, careers, and broad wage patterns.
- The companion paper focuses more directly on the firm's wage policy.
- Together, the two papers show that wages, promotions, and job assignments form a linked system.
- This paper provides the organizational map that makes the wage-policy analysis interpretable.

Interpretation: The paper's incremental role is to build the empirical anatomy of the firm: the hierarchy, flows, and career structure that underlie compensation dynamics.

3 Data and measurement

3.1 Firm and sample

- The data come from one large U.S. firm observed from 1969 to 1988.
- The analysis focuses on management employees.
- The sample contains job titles, salary, performance ratings, dates, and movements across jobs and out of the firm.
- The firm grows substantially over the sample period, especially in the 1970s and early 1980s.

Interpretation: Because the paper observes the same firm for a long horizon, it can separate stable organizational rules from temporary macro or firm-growth conditions.

3.2 Job titles and levels

- The authors infer a hierarchy from transitions among major job titles.
- Major titles are grouped into eight levels.
- Moves up the diagonal of the transition matrix are interpreted as promotions.
- Moves down are interpreted as demotions, while horizontal moves are lateral transfers.
- The hierarchy is remarkably stable even as the firm grows.

Interpretation: The level structure is not imposed by the researchers. It is inferred from actual career movements, which makes it a revealed organizational hierarchy.

3.3 Career outcomes

The main career outcomes include:

- entry into the firm;
- exit from the firm;
- promotion across levels;
- demotion or lateral transfer;
- time to promotion;
- salary level and salary growth;
- performance rating.

Interpretation: These outcomes allow the paper to ask whether careers look like internal job ladders, spot-market transactions, or a hybrid system.

3.4 Salary measurement

- Salaries are converted into real 1988 dollars.
- The paper studies mean salaries by level, salary ranges within levels, and salary changes after promotions.
- It also studies within-title salary quartiles to measure how salary position affects subsequent raises.

Interpretation: The salary data reveal both administrative wage structure and individual variation. Levels matter, but workers within the same title can earn very different salaries.

4 Empirical framework and key objects

4.1 Transition matrix among titles

Let T_{ab} denote the share of workers in old title a who appear in new title b the next year. The transition matrix summarizes movement as

$$T = [T_{ab}]_{a,b \in \mathcal{J}}, \quad (1)$$

where $\mathcal{J}$ is the set of major job titles.

- Diagonal entries represent staying in the same title.
- Entries above the diagonal represent promotions.
- Entries below the diagonal represent demotions.
- Off-diagonal entries within a level represent lateral transfers.

Interpretation: The transition matrix is the central empirical object for reconstructing the hierarchy. It transforms raw personnel records into an organizational map.

4.2 Promotion rates and time to promotion

For worker i in level ℓ , the promotion rate is

$$P_{i,t}^\ell = 1\{\text{Level}_{i,t+1} > \text{Level}_{i,t}\}. \quad (2)$$

The paper also computes the distribution of time until promotion or exit from a given level:

$$\Pr(\text{promotion by } s \mid \text{Level} = \ell), \quad \Pr(\text{exit by } s \mid \text{Level} = \ell). \quad (3)$$

Interpretation: Time-to-promotion patterns reveal whether jobs are true career ladders and whether some workers move on fast tracks.

4.3 Salary regressions

The wage analysis estimates regressions of the form

$$\log(\text{Salary}_{i,t}) = \alpha + \sum_{\ell} \beta_{\ell} 1\{\text{Level}_{i,t} = \ell\} + \gamma X_{i,t} + \varepsilon_{i,t}, \quad (4)$$

where $X_{i,t}$ includes tenure, performance, demographics, and other controls.

Interpretation: The size of the level coefficients shows how much salary is attached to hierarchical position, while residual variation shows how much pay differs within levels.

4.4 Green-card raise effects

The paper studies annual raises as a function of prior salary position within title and performance rating:

$$\text{Raise}_{i,t} = f(\text{SalaryQuartile}_{i,t-1}, \text{Performance}_{i,t-1}, \text{Level}_{i,t-1}) + u_{i,t}. \quad (5)$$

Interpretation: The “green card” idea is that administrative salary guidelines compress wages within jobs: workers high in the salary range receive smaller raises, while workers low in the range receive larger raises.

5 Identification and interpretation

5.1 What is identified by the data

- The data identify the firm’s actual internal hierarchy and career flows.
- They identify correlations between levels, promotions, salary, performance, and exits.
- They identify administrative regularities, such as salary ranges and promotion premia.
- They do not identify causal effects of promotions or salaries in a randomized sense.

Interpretation: This is a descriptive and theory-disciplining paper, not a natural experiment. Its strength is detailed measurement inside the firm.

5.2 Why one firm is still informative

- The paper does not claim that every firm has the same hierarchy or wage policy.
- The value of one-firm data is depth: a complete internal map is impossible in standard labour market surveys.
- The patterns are rich enough to challenge and refine theories of internal labour markets.

Interpretation: One firm cannot establish universal facts, but it can reveal mechanisms and constraints that aggregate data hide.

5.3 Main limitations

- The firm is anonymous, and the industry is not the main object of study.
- Some variables of theoretical interest, such as outside offers or firm-specific human capital, are not directly observed.
- Performance ratings are internal measures and may reflect managerial discretion.
- The analysis is primarily descriptive.

Interpretation: These limitations are real, but the paper's unusually rich personnel records make it foundational for personnel economics.

6 Tables and figures

6.1 Figure I: firm growth, wages, and turnover

Read:

- Panel A shows growth in assets, net income, and management employees from 1970 to 1988.
- Panel B shows mean real salary and return on assets.
- Panel C shows entry, exit, and employment growth.
- The firm expands substantially and experiences changing financial performance over the sample.

Economic message: The personnel system is studied in a growing and changing firm. The stability of the internal hierarchy is therefore not simply the result of a static environment.

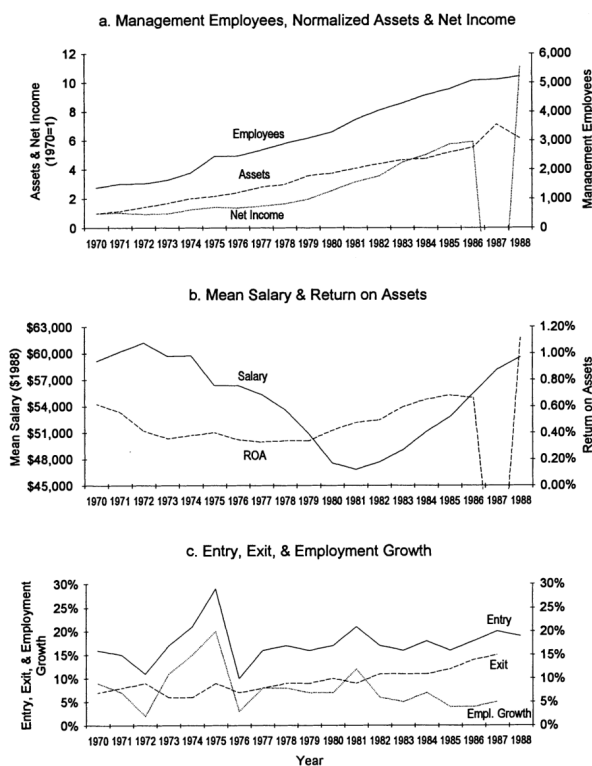


FIGURE I
 Managerial Employment, Financial Performance, Wages, and Turnover, 1970–1988
 All financial variables are calculated in 1988 constant dollars. In 1987, normalized net income = -13, and ROA = -1.1%.

expansion of the 1980s. Therefore, we observe the firm over a period of growth and varying economic conditions.

6.2 Table I and Figure II: hierarchy from transitions

Read:

- Table I is the transition matrix among major job titles from 1969 to 1988.
- The matrix is close to triangular, which makes it possible to infer a hierarchy of titles and levels.
- Figure II displays the implied hierarchy graphically.

- Titles are grouped into eight levels, with dense upward movements along a small number of promotion paths.

Economic message: The hierarchy is revealed by actual employee movements. This is direct evidence of an internal labour market with structured job ladders.

TRANSITION MATRIX BETWEEN MAJOR JOB TITLES, 1969-1988																								
vel		Old title	New title (next year)																Total %	Sample size				
			Level																					
			Exit	Ass't.	Asst.	K	H	M	L	K	J	I	H	G	F	Sr.	H	E			D	C	B	A
A	Entry		51	10	2	6	3	1	1	2	0	8	2	0	6	0	0	0	0	0	0	7	100	10,648
1	Asst. J	12	54	0	0	17	10	0	2	1	6	1	1	0	0	0	0	0	0	0	0	2	100	10,366
	Asst. K	7	1	74	4	4	8	6	0	1	0	1	0	0	0	0	0	0	0	0	0	0	100	4,075
	Asst. H	10	2	65	2	2	0	2	0	0	17	0	0	0	0	0	0	0	0	0	0	1	100	5,118
2	M	14	0	0	0	56	0	0	0	0	35	1	0	0	0	0	0	0	0	0	0	2	100	5,898
	L	7	0	0	0	1	74	2	0	0	0	17	0	0	0	0	0	0	0	0	0	2	100	5,110
	K	7	0	0	0	2	376	1	0	0	0	11	0	0	0	0	0	0	0	0	0	1	100	1,788
	J	12	0	0	0	2	2	0	64	0	18	0	0	0	0	0	0	0	0	0	0	1	100	917
	I	26	0	0	0	1	0	0	148	23	0	0	0	0	0	0	0	0	0	0	1	100	412	
	H	8	0	0	0	0	1	0	72	2	14	0	0	0	0	0	0	0	0	0	1	100	1,024	
	G	9	0	0	0	0	0	0	0	76	0	13	0	0	0	0	0	0	0	0	1	100	527	
	F	29	0	0	0	1	0	0	1	151	8	0	0	0	0	0	0	0	0	0	10	100	641	
	Sr. H	9	0	0	0	0	0	0	0	1	83	8	0	0	0	0	0	0	0	0	0	100	448	
	E	10	0	0	0	0	0	0	0	0	0	89	1	0	0	0	0	0	0	0	1	100	11,423	
	D	6	0	0	0	0	0	0	0	0	0	0	87	3	0	0	0	0	0	0	3	100	987	
	C	6	0	0	0	0	0	0	0	0	0	0	1	85	1	0	0	0	0	0	7	100	211	
	B	10	0	0	0	0	0	0	0	0	0	0	0	0	79	7	0	0	0	3	29	100	29	
	A	11	0	0	0	0	0	0	0	0	0	0	0	0	0	90	0	0	0	0	19	100	19	
	Other	15	2	0	0	2	1	0	0	0	0	2	1	0	4	1	0	0	0	0	72	100	3,385	
Total		9	16	6	1	9	8	3	1	1	19	1	1	18	2	0	0	0	0	6	69	100	69,840	

Shows all transitions between job titles, including entry, exit, and stays, from 1969 to 1988, as a percentage of movements from a title (or entry) to a new title (or exit) in the next year; other than major titles from Figure II are combined as "Other." Titles F and I were added in later years, so that their percentages are higher in those years than over the entire sample. Shaded cells show promotions of one level. Moves above and to the right of the shaded cells, except from Entry and to Other, are promotions of more than one level. Numbers in boxed cells w stays (the diagonals of the boxes) and lateral transfers (the off-diagonals). Moves below and to the left of the boxes, except from Other and to Exit, are demotions. Zero denotes empty cells that round to zero, while "-" denotes empty cells.

Table I: transition matrix between major job titles

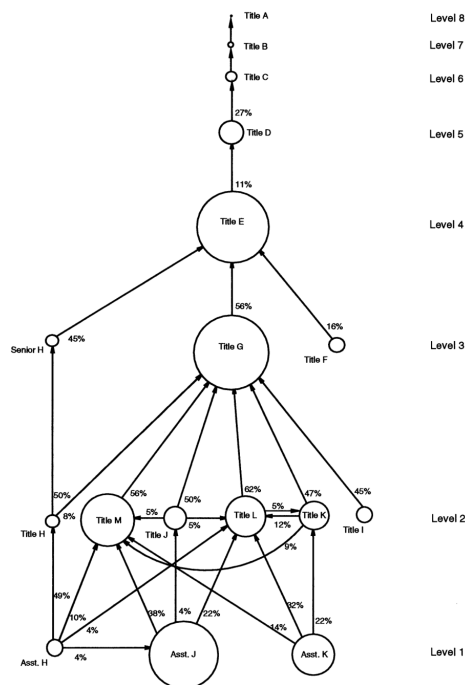


FIGURE II
Titles and Hierarchical Levels: 1970-1987

Circles represent all titles comprising at least one-half percent of person-years in the sample, except for Titles A (CEO), B, and C. The area of each circle represents the relative number of person-years for that title. For reference, Title G comprises 18 percent of person-years. Arrows show movements out of a title; numbers next to each arrow are the fraction of all movements out of the title (including exits) on that path. Because movements here exclude stays, these numbers represent *ultimate* (rather than annual) promotion rates. Only movements of 4 percent or greater out of a position, and no exits, are shown. Numbers for movements among the top three

Figure II: hierarchy of titles and levels

6.3 Figures III and IV: stability and promotion rates

Read:

- Figure III shows the relative size of levels in selected years.
- Despite firm growth, the basic hierarchy remains stable over time.
- Figure IV shows promotion rates from each level over adjacent periods.
- Promotion patterns differ across levels, but the same general hierarchy persists.

Economic message: The internal labour market is not a temporary artifact. It is a persistent organizational structure.

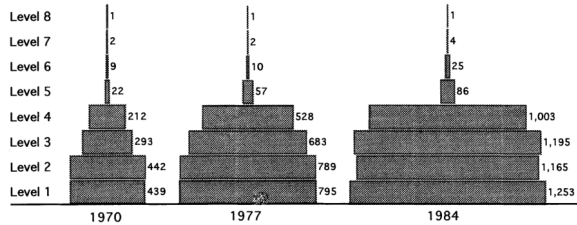


FIGURE III
Size of Levels in Selected Years

Each bar shows the relative size of Levels 1–8 that year. Numbers next to bars are the number of employees at that level that year.

seventeen titles? Equally interesting, why did the firm's structure not change over time and firm size? Theories of hierarchical structures and monitoring in hierarchies [Beckmann 1977; Calvo and Wellisz 1979] predict that levels expand proportionately with firm size, as we find. They also predict that the number of levels will grow with firm size, but that is not what we observe.

Steady State Job Transitions. The firm was stable in the

Figure III: level distribution over time

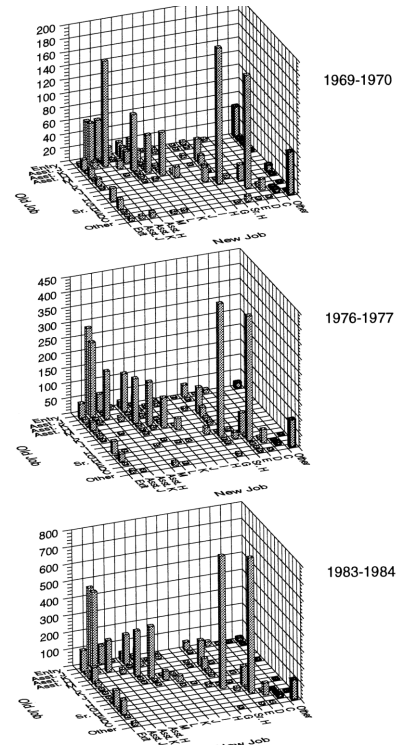


FIGURE IV
Job Transition Patterns for Major Titles in Selected Years
Each bar represents the number of transitions from a title in the previous year (Old Job) to a title in that year (New Job), including entry, stays, and exits. Jobs other than major titles from Figure II are combined as "Other."

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Figure IV: promotion rates over time

6.4 Tables II–IV: careers, entry, exit, and level changes

Read:

- Table II shows career entry, exit, and movement patterns by level.
- Entry and exit are not limited to a small number of ports, which challenges a classic internal labour market assumption.
- Table III compares subsequent careers of new hires and promotees.
- Table IV examines time to promotion and exit by level.
- Promotion rates vary substantially by level, while exit rates are more uniform than many theories predict.

Economic message: The firm has a strong internal labour market, but not the strict ports of entry and exit emphasized in classic theory. Promotions matter more than exits for sorting workers across levels.

TABLE II
CAREER AND LEVEL CHARACTERISTICS

	Level				
	1	2	3	4	5-8
Number of outside entrants at level	2,714	579	365	189	14
Percent with 1-year careers	10.7	15.2	10.7	15.3	7.1
Percent with 2-year careers	10.4	10.2	10.1	7.9	14.3
Percent with 5-10-year careers	25.5	19.7	25.5	24.9	42.9
Of which:					
Percent holding 1 title	16.0	25.4	38.7	66.0	66.7
Percent holding 2 titles	32.5	36.8	59.1	27.7	33.3
Percent holding 3 titles	34.3	35.1	2.2	4.3	0.0
Percent holding 4 titles	16.4	2.6	0.0	2.1	0.0
Percent holding 5+ titles	0.8	0.0	0.0	0.0	0.0
Percent with 10+ year careers	39.8	38.5	35.6	30.7	28.6
Of which:					
Percent holding 1 title	5.9	11.7	27.7	55.2	50.0
Percent holding 2 titles	16.0	18.8	59.2	34.5	0.0
Percent holding 3 titles	32.5	48.0	9.2	8.6	50.0
Percent holding 4 titles	37.0	17.5	3.1	1.7	0.0
Percent holding 5+ titles	8.5	4.0	0.8	0.0	0.0
Average age of new hires	32.5	35.3	38.3	43.5	43.4
Average years of schooling of new hires	15.6	16.4	16.5	17.0	18.0
Average work experience of new hires	10.9	12.9	15.8	20.5	19.4
Number promoted into level	0	1,617	1,098	589	87
Average age of those promoted into level	N/A	34.0	36.1	38.7	45.6
Average years of schooling of those promoted into level	N/A	15.7	16.1	16.5	17.1
Average work experience of those promoted into level	N/A	12.3	14.0	16.2	22.5
Number of person-years in level	16,981	17,725	17,253	13,892	1,643
Percent of all entrants into level who were outside hires	98.6	26.1	29.7	24.9	10.0
Promotion rate (percent) per year	26.3	18.0	10.9	1.4	4.5
Exit rate (percent) per year	11.4	11.5	11.0	9.6	8.2

The top panel of the table uses only those who entered the level from outside the firm between 1970 and 1979. The middle panel uses all those who entered the firm (at any level) between 1970 and 1979 and were subsequently promoted to the level at some time up to 1988. The bottom panel uses all employees at the firm in all years. Work experience is age - schooling - 6.

Ports of Entry and Exit. Table II provides direct evidence on a key assumption underlying ILM theory, that internal jobs are shielded from external labor market pressures except at ports of

TABLE III
CAREER ATTAINMENT, NEW HIRES VERSUS INCUMBENTS PROMOTED INTO
LEVELS 2-3, 1970-1979

Current level	Statistic	New hire/ incumbent	Level 2 entrants: years since entering level 2									
			1	2	3	4	5	6	7	8	9	10
1	% of Remaining	New hire		1.4	1.6	1.8	2.1	2.0	1.8	1.1	0.8	1.0
		Incumbent					0.1					
2	% of Remaining	New hire	100.0	79.4	51.5	39.7	33.3	29.0	24.6	23.3	22.5	22.0
		Incumbent	100.0	84.2	49.7	32.1	23.7	18.4	13.7	11.6	10.1	8.6
3	% of Remaining	New hire		19.0	45.2	47.5	39.0	34.9	31.8	29.4	25.4	24.4
		Incumbent		15.7	49.3	59.2	53.7	48.5	46.3	42.2	37.8	36.4
4	% of Remaining	New hire		0.2	1.6	10.9	25.6	33.9	41.1	44.7	48.4	46.8
		Incumbent		0.1	0.9	8.7	22.5	32.8	39.4	45.4	50.6	53.1
5-6	% of Remaining	New hire						0.3	0.7	1.5	2.9	5.9
		Incumbent						0.2	0.7	0.9	1.5	1.9
1-6	Level (average)	New hire	2.00	2.18	2.47	2.68	2.88	3.02	3.14	3.22	3.30	3.35
		Incumbent	2.00	2.16	2.51	2.77	2.99	3.15	3.27	3.36	3.44	3.49
	Level (variance)	New hire	0.00	0.18	0.31	0.48	0.66	0.72	0.74	0.74	0.77	0.85
		Incumbent	0.00	0.14	0.27	0.35	0.47	0.50	0.49	0.48	0.50	0.48
	Exit rate (%)	New hire	15.2	12.0	10.7	12.7	8.6	8.8	6.1	6.9	9.0	7.8
		Incumbent	11.3	11.1	9.1	7.2	8.7	8.5	7.2	11.3	10.0	7.5
	N	New hire	579	490	431	385	336	307	280	262	244	222
		Incumbent	1,346	1,194	1,062	964	894	816	747	692	613	550
Current level	Statistic	New hire/ incumbent	Level 3 entrants: years since entering Level 3									
			1	2	3	4	5	6	7	8	9	10
1-6	Level (average)	New hire	3.00	3.15	3.36	3.43	3.50	3.59	3.63	3.69	3.72	3.72
		Inc. (hired at Level 2)	3.00	3.05	3.37	3.50	3.61	3.73	3.81	3.88	3.94	4.07
		Inc. (hired at Level 1)	3.00	3.03	3.30	3.50	3.61	3.72	3.77	3.84	3.87	3.93
	Level (variance)	New hire	0.00	0.14	0.28	0.31	0.34	0.33	0.33	0.35	0.34	0.38
		Inc. (hired at Level 2)	0.00	0.06	0.25	0.29	0.32	0.27	0.31	0.35	0.35	0.35
		Inc. (hired at Level 1)	0.00	0.03	0.21	0.26	0.26	0.25	0.24	0.23	0.21	0.25
	Exit rate (%)	New hire	10.7	11.3	12.2	11.9	14.3	11.5	8.9	6.5	9.7	8.1
		Inc. (hired at level 2)	8.4	11.2	8.7	6.6	10.3	7.1	7.0	8.3	10.0	13.0
		Inc. (hired at Level 1)	8.2	8.4	8.7	8.8	6.3	8.7	9.2	11.8	9.5	11.5
	N	New hire	365	326	288	253	223	191	169	154	144	130
		Inc. (hired at Level 2)	225	206	183	167	156	140	129	120	110	99
		Inc. (hired at Level 1)	547	502	459	419	382	358	327	297	262	236

The top panel shows progressions of all employees entering Level 2 in 1970-1979, either new hires or incumbents promoted from Level 1. % of Remaining is the percentage of new or incumbent employees remaining in the firm who are in the given level. The middle panel shows the exit rate, and average and variance of the level attained by remaining new hires or incumbents, by number of years in Level 2. The bottom panel shows the same statistics for new hires or incumbents (promoted from either Level 1 or 2) in Level 3. Exit rates may be slightly smaller than implied by sample sizes, as a few employees moved to jobs we were unable to assign to a level. No employees in these groups were promoted to Levels 7-8. Inferences do not change if we restrict the sample to U. S. employees.

Table III: new hires versus promotees

TABLE IV
TIME TO PROMOTION IN LEVEL 1 VERSUS TIME TO PROMOTION OR EXIT IN LEVEL 2

Years at Level 1	Statistic	Years at Level 2 before promotion or exit									
		1	2	3	4	5	6	7	8	9	10+
1	Promotion rate (%)	22	40	29	23	11	18	14	6	25	7
	Exit rate (%)	16	18	16	10	15	12	11	12	8	11
	N	1494	851	313	140	80	51	28	17	12	28
2	Promotion rate (%)	14	27	25	21	14	14	12	8	15	7
	Exit rate (%)	10	10	9	9	7	14	7	11	10	15
	N	1134	805	468	272	164	105	58	36	20	27
3	Promotion rate (%)	6	19	17	23	17	20	14	9	0	4
	Exit rate (%)	9	8	9	5	11	14	8	14	18	18
	N	485	393	248	168	115	64	37	22	11	28
4	Promotion rate (%)	5	17	18	14	16	20	4	13	13	17
	Exit rate (%)	7	10	5	4	7	15	8	13	25	8
	N	224	175	120	78	56	41	24	15	8	12
5	Promotion rate (%)	3	9	10	4	18	10	8	10	0	0
	Exit rate (%)	9	7	10	12	6	10	0	20	0	8
	N	114	97	70	51	34	20	13	10	6	13
6	Promotion rate (%)	8	11	24	22	10	25	20	0	0	0
	Exit rate (%)	8	11	3	17	0	0	0	0	0	0
	N	53	44	29	18	10	8	5	4	2	6
7+	Promotion rate (%)	1	6	19	6	7	0	7	0	0	0
	Exit rate (%)	8	11	3	17	0	0	0	0	0	0
	N	78	64	48	35	29	21	15	6	4	4

Table IV: time to promotion and exit

6.5 Figures V-VI and Table V: salary structure

Read:

- Figure V shows mean real salary by level from 1969 to 1988.
- Higher levels have much higher pay, and the level-pay gradient is steep.
- Figure VI shows large salary ranges within levels and substantial overlap across adjacent levels.
- Table V confirms that levels explain much of pay, but that substantial within-level variation remains.

Economic message: Pay is attached to hierarchy, but the wage system is not a pure job-rate system. The firm uses both job levels and individual salary variation.

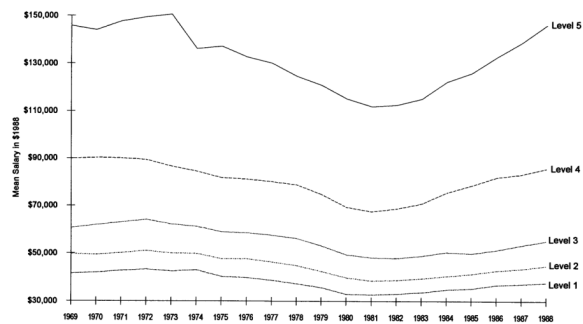


FIGURE V
Mean Salary by Level, 1969–1988
Calculated in 1988 constant dollars.

Panel a shows all levels, and Panel b shows greater detail for Levels
Figure V: mean salary by level

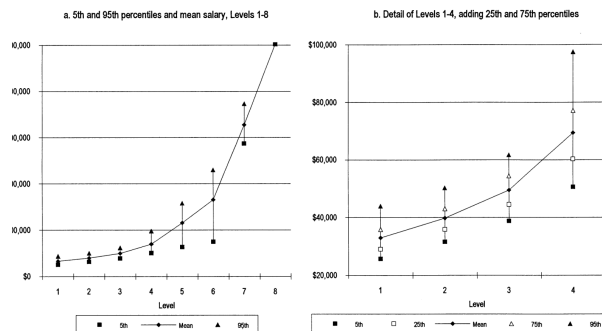


FIGURE VI
Salary Ranges by Level, 1980
Calculated in 1988 constant dollars. Level 8 has only one employee.

Figure VI: salary ranges within levels

TABLE V
EFFECTS OF HUMAN CAPITAL AND HIERARCHICAL LEVEL ON CURRENT SALARY

Independent variables	1970–1988 Pooled regressions			1985 cross section
	i. Human capital	ii. Levels	iii. Combined	
Year dummies	yes	yes	yes	no
Sex dummy	yes	no	yes	yes
Race dummy	yes	no	yes	yes
Intercept	10.53 (.02)	10.58 (.01)	10.59 (.01)	10.53 (.01)
13–16 years education	.04 (.004)	.	-.01 (.002)	-.01* (.008)
17–18 years education	.24 (.004)	.	.08 (.008)	.09 (.01)
19+ years education	.34 (.008)	.	.11 (.006)	.10 (.02)
Tenure	.05 (.001)	.	-.003 (.0009)	-.01 (.003)
Tenure ²	-.001 (.00008)	.	.0004 (.00006)	.0008 (.0002)
Level 2	.	.16 (.002)	.14 (.002)	.16 (.01)
Level 3	.	.36 (.003)	.33 (.003)	.33 (.01)
Level 4	.	.74 (.003)	.68 (.003)	.74 (.01)
Level 5	.	1.16 (.01)	1.08 (.01)	1.13 (.03)
Levels 6–8	.	1.43 (.02)	1.35 (.02)	1.49 (.05)
R ²	.35	.68	.71	.70
R ² , regression w/out year dummies	.30	.64	.67	N/A
N	43,629	43,629	43,629	3955
Dependent mean	10.75	10.75	10.75	10.76

In all regressions the dependent variable is the log of real salary in 1988 constant dollars. Education variables are dummies for those levels of schooling; BA = 16, etc. Tenure is years with company since hire. Regressions similar to i–iii using fixed effects rather than sex, race, and education dummies do not change the results significantly. Standard errors are in parentheses. Due to large sample sizes, most coefficients are significant at the 1 percent level. The one that is not is indicated by an asterisk.

among members of the same entry cohorts, and across titles. Wages are not determined solely by job or level.

At the same time, levels are important for wage determination. This is quantified in Table V, which presents wage regressions

6.6 Tables VI–VIII: promotion premia and salary selection

Read:

- Table VI reports salary premia associated with staying, demotion, and promotion.
- Promotion premia rise with level and explain part of the wage gap across levels.
- Table VII shows where promotees come from in the prior salary distribution and where they land in the new salary distribution.
- Promotees tend to come from higher pay deciles in the old title and enter lower deciles in the new title.
- Table VIII shows exit rates by salary decile and level.
- Exit rates vary surprisingly little across salary deciles, weakening a simple selection-by-exit interpretation.

Economic message: Salary selection into levels is driven mainly by promotion, not exit. Promotions help sort high-performing or high-salary employees upward while maintaining salary ranges within titles.

Table VII shows where promotees came from in the pay distribution in their prior job and where they went to in the pay distribution in their new job. The table presents the distribution of promotees across salary deciles in prior jobs, as well as the

TABLE VI
SALARY PREMIUMS BY TYPE OF JOB TRANSITION, AND ACROSS LEVELS

Level	% salary premiums on:			% difference, mean pay vs. level below
	Stay	Demotion	Promotion	
1	−0.5%	−0.7%	.	.
2	−0.4	−0.2	5.1	18
3	0.1	−3.2	5.6	23
4	0.8	0.4	7.4	47
5	−0.1	0.5	8.7	64
6	0.1	.	4.5	40
7	−0.9	.	22.3	107
8	0.0	.	14.8	48
1–8	0.0%	−0.9%	5.8%	

Salary premiums are mean percentage raises (across years) for those with that type of job move, relative to the mean raise (that year) of all employees in the firm who did not change level. Statistics are shown by current level, not level prior to promotion or demotion. % difference, mean pay vs. level below is the mean (across years)

Table VI: promotion premia

TABLE VII
DISTRIBUTION OF PAY FOR PROMOTEEES IN SALARY DECILES BEFORE AND AFTER PROMOTION

		Percentage in each salary decile											
Promotion	Statistic	N	Bottom	2nd	3rd	4th	5th	6th	7th	8th	9th	Top	All
Level 1 to Level 2	Decile before promotion	3947	4.4%	7.7%	9.1%	9.6%	9.3%	10.6%	12.6%	11.2%	13.1%	12.5%	100%
	Decile after promotion	3947	17.1	15.4	13.1	10.6	8.3	8.8	7.1	7.3	7.1	5.2	100
Level 2 to Level 3	Decile before promotion	2910	2.7	5.1	7.2	8.3	9.7	11.7	12.1	13.9	14.3	15.1	100
	Decile after promotion	2910	18.4	19.1	15.1	11.9	8.6	7.0	5.6	5.5	5.7	3.0	100
Level 3 to Level 4	Decile before promotion	1666	2.0	3.4	5.3	7.7	6.8	12.9	12.7	13.5	15.4	20.2	100
	Decile after promotion	1666	32.7	23.9	15.7	10.6	5.9	5.9	1.7	1.6	1.2	0.7	100
Level 4 to Level 5	Decile before promotion	165	3.6	1.8	0.6	1.2	0.8	12.1	2.4	8.5	18.8	50.3	100
	Decile after promotion	165	7.9	25.5	23.0	15.8	9.1	10.9	3.6	1.8	1.8	0.6	100

Shows distribution of salary deciles out of which employees were promoted, and into which they were promoted, for those promoted one level at a time out of Levels 1 to 4. Salary deciles recalculated over all employees in the same title in the relevant year.

Table VII: promotee salary distributions

TABLE VIII
EXIT RATES BY LEVEL AND SALARY DECILE

Level	N	Percentage in each salary decile										χ^2 Stat
		Bottom	2nd	3rd	4th	5th	6th	7th	8th	9th	Top	
1	15,817	13.9%	13.4%	11.1%	11.3%	12.7%	10.6%	9.1%	11.7%	10.5%	10.7%	26.8*
	16,129	12.8	12.2	11.4	9.9	11.7	12.4	11.3	12.0	11.8	11.4	7.7
	15,140	11.7	11.1	10.8	11.8	10.0	10.4	8.8	11.0	12.0	11.2	11.7
	11,932	12.1	10.7	8.0	7.4	11.1	9.2	9.0	11.7	10.2	10.1	25.1*
	1,078	11.4	11.2	4.1	11.7	11.2	7.1	9.7	2.1	8.7	10.3	11.5
	407	9.1	11.4	9.4	6.5	12.1	8.8	7.3	0.0	6.3	0.0	6.8
2	60,503	12.6	11.9	10.4	10.2	11.4	10.6	9.6	11.4	11.1	10.8	37.44*

Annual percentage exit rates are shown. Deciles are calculated within job title and year. The last column shows Chi-square statistics for the hypothesis that exit rates for all deciles in a are the same. An asterisk indicates significant between-cell differences at the 1 percent level.

6.7 Table IX: green-card effects on raises

Read:

- Table IX shows percentage real raises by salary quartile and performance rating.
- Within a level and performance rating, workers in lower salary quartiles often receive larger raises.
- Workers at the top of the salary range receive smaller raises, consistent with administrative pay guidelines.
- The pattern holds across multiple levels.

Economic message: Administrative salary rules compress pay within jobs. The firm does not simply pay workers their current marginal product; it manages salary ranges through raise guidelines.

TABLE IX
“GREEN CARD” EFFECTS OF RELATIVE SALARY IN TITLE ON RAISES, BY
PERFORMANCE RATING

Level	Salary quartile in title	Performance rating				
		1	2	3	4–5	1–5
1	Top	3.2%	2.5%	1.8%	–1.6%	2.3%
	3rd	4.3	3.1	1.3	–2.7	2.5
	2nd	4.7	3.8	1.4	–2.5	3.0
	Bottom	7.2	5.5	2.8	–1.1	4.8
	N	1067	3001	2232	101	6401
2	Top	6.4	3.7	1.8	0.4	3.9
	3rd	7.2	4.5	1.5	–2.8	4.3
	2nd	8.7	5.2	2.0	–1.6	5.1
	Bottom	10.4	6.6	1.9	–2.5	6.0
	N	2021	5371	2202	83	9677
3	Top	6.6	3.9	1.6	–1.9	4.5
	3rd	7.5	4.7	1.5	–2.7	5.0
	2nd	8.3	5.7	1.7	–1.4	5.7
	Bottom	10.1	6.3	2.1	–1.5	6.4
	N	2729	5073	1421	67	9290
4	Top	7.3	4.2	0.9	–3.8	5.6
	3rd	8.0	4.9	1.5	–1.9	6.0
	2nd	8.7	5.0	1.8	–2.0	6.4
	Bottom	9.4	6.4	1.8	–3.2	7.1
	N	3398	3362	607	33	7400
1–7	Top	6.6	3.7	1.7	–1.3	4.3
	3rd	7.3	4.3	1.4	–2.5	4.5
	2nd	8.1	5.0	1.7	–2.0	5.1
	Bottom	9.4	6.2	2.2	–1.8	6.1
	All	7.7%	4.8%	1.8%	–1.9%	5.0%
	N	9679	17,320	6708	308	34,015

Shows the mean percentage raise, in 1988 constant dollars, by performance rating given in the same period. Statistics are broken out by current level and by the previous year’s salary quartile among those in the same title. Levels 5–7 are not shown individually because of small sample sizes. Level 8 employees never received performance ratings. Sample sizes for each level/rating combination are also shown. Sample sizes for specific quartiles are approximately one fourth of these numbers.

7 Short summary

- The paper analyzes twenty years of personnel data from one large firm.
- The firm has a clear and stable hierarchy of job titles and levels.
- Career movements reveal structured promotion paths and fast tracks.
- Promotions are central to career progress and pay growth.
- Exit rates vary surprisingly little with tenure, salary, and level.
- The firm has many features of an internal labour market but not clean ports of entry and exit.

- Salary is strongly related to level, but there is substantial within-level salary dispersion.
- Promotion premia explain part of the pay gap across levels.
- Administrative rules compress wages within titles through salary-quartile-based raises.
- The paper is foundational because it shows how personnel records can be used to study the internal economics of firms.

8 Conclusion

8.1 Core conclusion

Baker, Gibbs, and Holmstrom show that a firm’s internal labour market can be mapped quantitatively using personnel records. The firm has a clear hierarchy, structured career paths, strong promotion dynamics, and administrative salary rules. These features closely resemble classic internal labour market descriptions, but the underlying theoretical explanation is more complicated than the classic account suggests.

8.2 Economic intuition

The firm creates careers rather than merely buying labour in spot markets. Employees move through a stable hierarchy, and pay depends heavily on level and promotion history. At the same time, individual variation within levels, salary-range rules, and weak exit patterns show that internal labour markets are not simple job-rate systems or pure firm-specific human-capital arrangements.

8.3 Takeaway

The paper’s enduring contribution is methodological and substantive. Methodologically, it demonstrates the value of administrative personnel data. Substantively, it shows that internal labour markets are structured, rule-governed systems in which jobs, promotions, salary ranges, performance ratings, and careers interact.